

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

April 2013

EE 206 Digital Electronics & Microprocessor

Date: 26.04.2013, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain Full adder circuit diagram using two half-adder circuit with truth table. [04]

(b) State and Prove commutative, associative and distributive laws of Boolean algebra. [03]

Q - 2 (a) Do as directed [04]

i) Convert $(B9F.AE)_{16}$ to octal

ii) Convert $(5497)_{10}$ to binary

iii) Convert $(101111010001)_2$ to decimal

iv) Convert $(5C7)_{16}$ to decimal

(b) Write note on 3 to 8 line decoder. [04]

(c) Prove that [04]

(i) $AB + AB'C + BC' = AC + BC'$

(ii) $[(AB)' + A' + AB]' = 0$

(d) Realize X-OR operation using Only NAND gates [02]

OR

Q - 2 (a) With the help of a logic diagram and a truth table, explain an Octal to Binary Encoder [06]

(b) Explain Diode Transistor Logic [04]

(c) With the help of neat circuit diagram, explain the working of a two-input CMOS NAND gate. [04]

Q - 3 (a) Simplify the following expressions using Karnaugh Map $\Sigma m(1,3,5,7,8,9,12,13)$ and implement it in universal logic.

[06]

(b) Write notes on level Triggered D Flip-Flop.

[04]

(c) (i) VLSI is used for integrating more than _____ components.

[04]

(ii) A multiplexer is also known as _____.

(iii) _____ circuit needs a memory element.

(iv) In the _____ shift register the data moves in both the directions.

OR

Q - 3 (a) Write a notes on BCD to 7 segment decoders.

[06]

(b) Give the comparison between Registers and Counters

[04]

(c) Write notes on 2-bit Synchronous Up counters using J-K Flip-flop with timing diagram.

[04]

SECTION - II

Q - 4 (a) What is flag? Draw the format of 8085 flag register and explain in detail.

[05]

(b) Explain Program Counter and Stack Pointer in 8085 microprocessor.

[02]

Q - 5 (a) Draw the complete pin diagram of 8085 microprocessor and explain the function of ALE and TRAP pin.

[07]

(b) Explain 8085 De-multiplexed Address and Data Bus with control signals with necessary diagram.

[04]

(c) Explain Instruction word size with example.

[03]

OR

Q - 5 (a) Explain the function of following instructions.

[07]

i) XRA r

ii) STA addr

iii) DAD rp

iv) XCHG

v) CMP M

vi) MVI M, data

vii) LHLD addr.

(b) Give the comparison between CALL and RET & PUSH and POP.

[04]

- (c) Calculate the number of memory chips needed to design 32K byte memory if the available chip size is 1024×1 . [03]

Q - 6 (a) Write an assembly language program for exchange the contents of memory location 3000H and 5000H. [03]

- (b) Write an assembly language program to find smallest no. in a data array. The number in a series are : 49, 60, 80. [03]

(C) Why data bus is called bidirectional? [02]

(d) Answer the following questions [06]

- (i) If the 8085 has fetched the machine code located at the memory location D00A H, specify the contents of the program counter.
- (ii) If the clock frequency is 5MHz, how much time is required to execute an instruction of 18 T-States?
- (iii) Write instruction to load 00H in the accumulator.
- (iv) Write the machine code for the instruction MOV H,A if the opcode = 01_2 , the register code for H = 100_2 and the register code for A = 111_2
- (v) Mention the categories of instruction and give two examples for each category.
- (vi) Which interrupt is highest priority and Non-Maskable interrupt in 8085 microprocessor?

OR

Q - 6 (a) Write a program to count from 0 to 9 with a one-second delay between each count. At the count 9, the counter should reset itself to 0 and repeat the sequence continuously. Use register pair HL to setup the delay and display each count at one of the output ports. Assume clock frequency of microcomputer is 1 MHz. [07]

(b) Write a program for a memory block starting from address D000H. Store 10 data bytes sequentially. Store the data in the same memory block in reverse order. [04]

(c) Write a program for add two 16-bit nos:- The first no. is stored at memory address D000 & D001, Second no. is stored at memory address D002 & D003. After add, stored the answer in memory address D004 & D005. [03]
